HOOST: A novel hyperplane-oriented over-sampling technique for imbalanced fault detection of aero-engines

Abstract

In general, training fault samples of aero-engines are very rare and only collected under one or a few operating conditions. However, due to diverse operating conditions and fault severities, testing fault samples may have higher diversity and larger distribution region than training fault samples, leading to a high missing detection rate. To address this issue, a hyperplane-oriented over-sampling technique (HOOST) is developed to synthesize training fault samples with higher diversity and larger distribution region. To be specific, HOOST not only adopts the interpolation strategy to synthesize in-distribution samples, but also adopts the extrapolation strategy to synthesize out-of-distribution samples. Moreover, the sampling factors in the extrapolation strategy are automatically set under the guidance of the initial classification hyperplane, rather than randomly selected, in order to improve the reasonableness of synthetic samples. Finally, the developed HOOST is integrated with a self-attention encoder-decoder and demonstrated through a series of experiments on an actual aero-engine dataset.

Keywords:

Aero-engines, data imbalance, over-sampling, out-of-distribution samples, fault detection

1. Introduction

The health condition of aero-engines has a decisive effect on flight safety. As a core component of aircraft operating in extremely harsh environments for long periods, aero-engines inevitably encounter some safety hazards (e.g., component fatigue, corrosion, and performance degradation). If these safety hazards are not detected and eliminated in a timely manner, they may be gradually deteriorating and lead to unplanned maintenance costs and serious human

casualties. Therefore, it is crucial to explore an accurate fault detection framework to detect the health condition of aero-engines [1]-[4].

In recent years, machine learning-driven fault detection methods for complex equipment have received increasing attention from researchers [5]-[7]. The major superiority of machine learning methods is that they can avoid complex professional analysis [8]. For example, Shang et al. apply the local window Transformer network to extract weak damage information and the graph convolution network to aggregate label semantic information, then fuse them to realize the damage detection of the aero-engine blade. Quan et al. [9] combine the deep dilated causal convolutional neural network and logistic regression to construct a stall fault detection framework for the axial compressor of aero-engines. These machine learning-based fault diagnosis methods require that the number of samples from different categories is statistically equivalent [10]. Hence, for gas turbines, they still confront the following challenges:

(1) **Class imbalance:** In practice, aero-engines are designed to be highly reliable, operate in the healthy state for a long period and occasionally malfunction. Training healthy samples (i.e., majority class samples) are very abundant, while training fault samples (i.e., minority class samples) are very time-consuming and rare. That is, minority class samples are much lower than majority class samples in number.

(2) **Distribution shift:** Training healthy samples are collected for long time, which can cover most operating conditions. Testing healthy samples may have similar distribution to training healthy samples, as depicted in Fig.1(a)-(b). In contrast, training fault samples only cover one or a few specific operating conditions. Due to diverse operating conditions and fault severities, testing fault samples may have different distribution region with training fault samples [11], [12], as depicted in Fig.1(a)-(b). In these cases, the decision hyperplane in Fig.1(b) (denoted as the initial decision hyperplane) may be biased towards the majority class. That is, the decision hyperplane tends to assign minority class samples to an incorrect category [13], [14], thus leading to a high missing detection rate.

Aiming at the aforementioned issues, over-sampling techniques have been considered promising candidates [15]. They attempt to correct the biased learning of the decision hyperplane by generating a number of minority class samples. For example, Chawla et al. [16] propose a popular Synthetic Minority Over-sampling TEchnique (SMOTE) and generate synthetic samples by interpolating inside the connection between each minority class sample and its nearest neighbors. Synthetic samples re-balance the data distribution and lead the distribution of minority class samples to become like that shown in Fig.1(c). For some classifiers (such as support vector machines), the decision hyperplane after SMOTE may not move; nevertheless, for other general classifiers, the decision hyperplane after SMOTE in Fig.1(d) may move towards the majority class samples, which is more desirable than the initial

decision hyperplane in Fig.1(b).

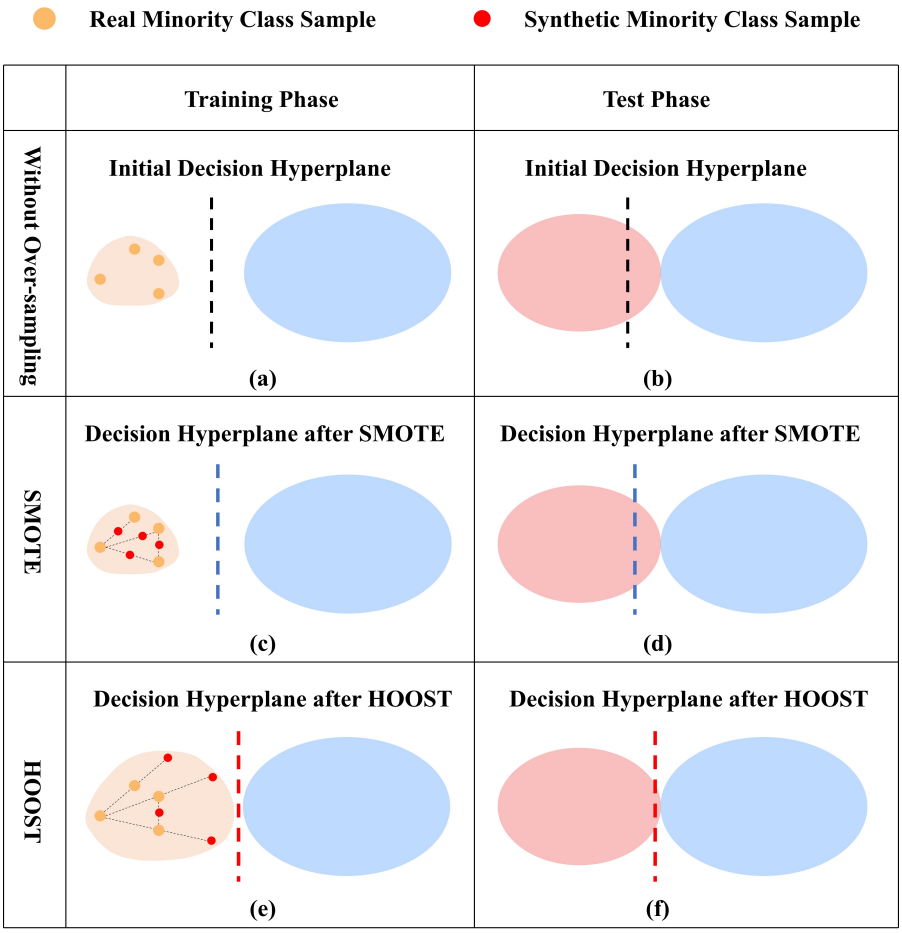


Fig.1 2D distributions and decision hyperplanes: (a)-(b) the condition without over-sampling, (c)-(d) the condition after SMOTE, and (e)-(f) the condition after HOOST, where represents the distribution region of (real and synthetic) training minority class samples, represents the distribution region of testing minority class samples, and represents the distribution region of training (or testing) majority class samples.

More notably, due to the fact that the interpolation strategy, the SMOTE can only synthesize in-distribution samples, where the in-distribution sample represents the sample that is located inside the distribution region of minority class samples in the training set. The introduction of in-distribution samples cannot expand the distribution region of training minority class samples, and the difference of the distribution region between training and test minority class samples hasn't become small, as depicted in Fig.1(c)-(d). Machine learning methods excel at learning a decision hyperplane on the training set, which can provide correct and confident categories for training samples and their extremely similar testing samples. When evaluated on test samples different from training samples, the decision hyperplane may have poor performance, as shown in Fig.1(d). Especially for test minority class samples that are close to training majority class samples, they may fall to the right side of the decision hyperplane (i.e., misclassified distribution region), thus still leading to misclassifications as majority class samples.

In this paper, a hyperplane-oriented over-sampling technique, namely HOOST, is developed. As depicted in Fig.1(e), the HOOST, oriented by the initial decision hyperplane, synthesizes not

only in-distribution samples but also out-of-distribution samples, where the out-of-distribution sample represents the sample that is located out of the distribution region of minority class samples in the training set. The introduction of out-of-distribution samples can expand the distribution region occupied by training minority class samples, thus making the disparity in the distribution regions of training and test samples become smaller, as depicted in Fig.1(e)-(f). Obviously, the decision hyperplane after HOOST in Fig.1(f) reduces the risk of misidentification of minority class test samples and is more desirable than that after SMOTE in Fig.1(d). Therefore, the afore-depicted synthetic out-of-distribution samples are conducive to improving the generalization ability of machine learning methods and are crucial for establishing the decision hyperplane.

Furthermore, the developed HOOST is applied in the feature space learned by a self-attention Encoder-Decoder, rather than the original space. In real-world scenarios, majority class samples often highly overlap with minority class samples. As a result, the over-sampling technique may generate noisy samples which belong to the minority class but distribute closer to the majority class samples. The introduction of noisy samples further complicates the identification of both categories of samples [17]. Herein, this paper leverages a self-attention Encoder-Decoder to map original samples to a feature space, where different categories of samples are distant, while the same categories of samples are close together. Subsequently, the HOOST adaptively synthetizes new samples in the learned feature space.

The main contributions of this paper are summarized below:

1) A hyperplane-oriented over-sampling technique is developed to generate new minority class samples in the learned feature space. Oriented by the initial decision hyperplane, an attention block is designed in the HOOST to learn sampling factors that can generate not only in-distribution samples but also out-of-distribution samples, thus reducing the misidentifications of unseen minority class samples.

2) A self-attention Encoder-Decoder is used to learn the well-separable feature space, which reduces the overlap of samples from different categories and decreases the possibility of synthesizing noisy samples.

3) A class-rebalanced and space-converted fault detection framework is presented to detect the health condition of aero-engines. The framework is evaluated using a real-world aero-engine dataset from an Asian airline.

The remaining sections are listed below. Section 2 reviews some related works. Section 3 illustrates the preliminary. Section 4 elaborates on the developed HOOST and its research motivation. Section 5 introduces the fault detection framework for aero-engines. Section 6 discusses the experimental results. Finally, Section 7 summarizes the main conclusions.

2. Related Works

In the literatures, the popular data re-sampling techniques can be mainly divided into three categories, including under-sampling, over-sampling, and hybrid re-sampling techniques [18], [19]. First, under-sampling techniques attempt to discard some majority class samples from the training set. For example, based on an unsupervised clustering algorithm, Lin et al. [20] propose a concise and efficient under-sampling technique that removes majority class samples in the edge region according to the distance from the cluster center. Goyal [21] develops a novel neighborhood-based under-sampling technique called N-US, which eliminates majority class samples having multiple memberships. Obviously, removed samples contain potentially useful information, thus negatively impacting the performance of the diagnostic framework.

Likewise, over-sampling techniques attempt to replicate or synthesize some minority class samples. For example, Bunkhumpornpat et al. [22] present the Safe-level SMOTE to carefully synthesizes minority class samples in the safe distribution region, in order to reduce the possibility of synthesizing noisy samples. Douzas et al. [23] propose a novel SMOTE-based over-sampling technique (namely KMeans-SMOTE) that employs the K-Means clustering approach to divide original data into different clusters and only applies SMOTE to synthesize minority class samples in safe clusters, thus avoiding generating cross-cluster noisy samples. Nevertheless, under the condition of the high inter-class overlap, over-sampling techniques are likely to generate noisy samples that will mislead the learning of decision hyperplanes.

In contrast, hybrid re-sampling techniques are an organic combination of under-sampling and over-sampling. On one hand, compared to under-sampling techniques, they can reduce the loss of useful healthy information. On the other hand, compared to over-sampling techniques, they can decrease the possibility of synthesizing noisy samples. For example, Ramentol et al. [24] utilize the improved SMOTE to synthesize minority class samples and adopt the rough set theory to remove noisy samples. Zhang et al. [25] propose an efficient hybrid re-sampling technique by combining a reverse k-nearest neighbors-based under-sampling and a SMOTE-based over-sampling technique to balance datasets, thus yielding accurate and robust classification results. It is notable that although the developed HOOST is an over-sampling technique, it can be combined with under-sampling techniques to serve as hybrid re-sampling method as well.

3. Preliminary

The objective of the implementation of the SMOTE is to re-balance the numbers of samples from minority and majority classes. The central idea of the SMOTE is to generate new samples by stochastically interpolating inside the line segments between neighboring minority class

samples, rather than randomly duplicating minority class samples, as depicted in Fig.2(a). To be specific, suppose that z is a minority class sample, and $\tilde{z}$ is one of its k nearest neighbors. First, the difference between z and $\tilde{z}$ is calculated. Then, this difference is multiplied by a sampling factor λ that obey the Uniform Distribution $\mathrm{U}(0,1)$, and is added with z. At last, the calculation formula of the synthetic sample (z_{new}) is $z_{\mathrm{new}} = z + \lambda(\tilde{z} - z)$. Moreover, Fig.2(b) shows that all possible positions of the synthetic sample are inside the connection line between z and $\tilde{z}$, and the specific position is determined by the sampling factor λ. For example, when $\lambda = 0.5$, $z_{\mathrm{new}} = 0.5 \times z + 0.5 \times \tilde{z}$ (i.e., the synthetic sample is located at the midpoint of the connection line).

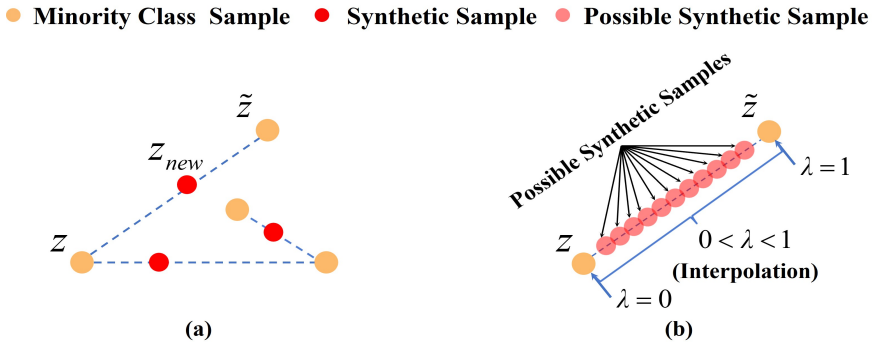


Fig.2 (a) The sketch of the SMOTE, (b) the interpolation strategy in the SMOTE.

4. Developed HOOST for Over-sampling

In this section, we illustrate the research motivations and elaborate on the developed hyperplane-oriented over-sampling technique.

4.1. Research motivations

The over-sampling technique is an important way to resolve the learning bias of classifiers under the condition of class imbalance. However, conventional over-sampling techniques (i.e., SMOTE, Safe-level SMOTE, KMeans-SMOTE, etc.) still face the challenging problem that only generates in-distribution samples when re-balancing minority and majority classes. As shown in Fig.1(c), the introduction of in-distribution samples cannot expand the distribution region occupied by minority class samples, whose gain to the estimation of the decision hyperplane is limited. This is due to the fact that the difference of the distribution region between training and test minority class samples hasn't become small, and the decision hyperplane after over-sampling may misclassify minority class test samples that fall outside the distribution region of training minority class samples, as shown in Fig.1(d). Therefore, compared to the generation of in-distribution samples, the ability to generate reasonable out-of-distribution samples is of more critical significance for over-sampling techniques.

4.2. Developed HOOST for synthetizing minority class samples

For the past several decades, over-sampling techniques (especially the SMOTE and its variants) have frequently been employed as a critical stage in many imbalanced classification tasks. In general, the in-distribution samples synthesized by these techniques cannot expand the distribution region of the minority class toward that of the majority class. To ensure a good performance in over-sampling, a key task is to generate out-of-distribution samples that are more crucial for establishing the decision hyperplane. The developed HOOST adopts not only the interpolation strategy but also the extrapolation strategy, which provides a promising way to accomplish this task.

The central idea of the HOOST is to synthesize out-of-distribution samples by adaptively sampling outward the line among neighboring minority class samples, as depicted in Fig.3(a). In the process of training, sampling factors can dynamically adjust the position of synthetic samples, thus making them gradually close to the initial decision hyperplane, so as to simulate the possible fault data distribution region more realistically. Moreover, Fig.3(b) shows that all possible positions of the synthetic sample are along the connection between z and $\tilde{z}$, and the specific position is determined by the sampling factor $\hat{\lambda}$.

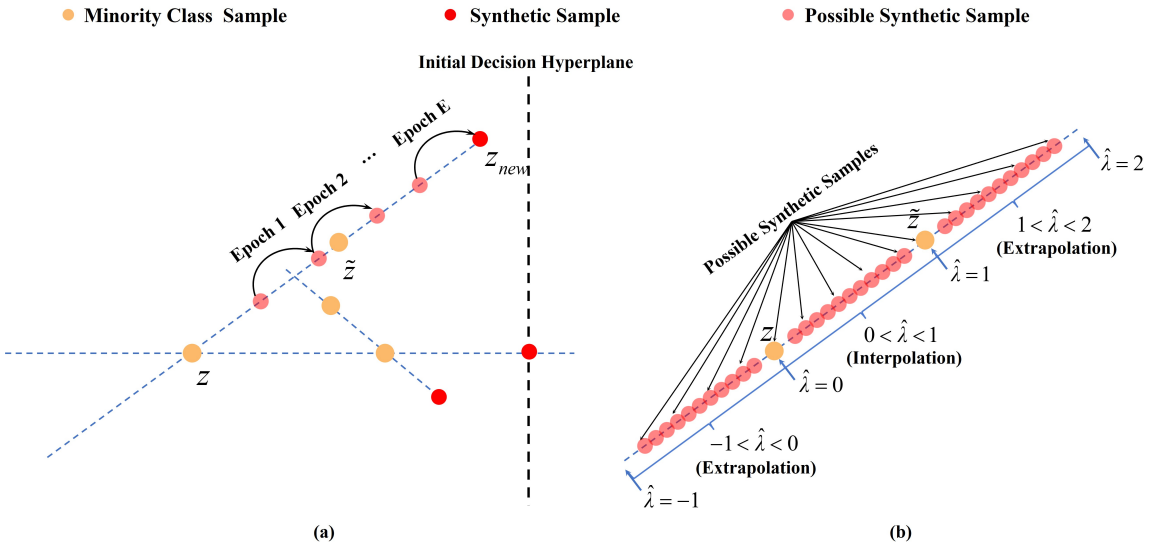


Fig.3 (a) The sketch of the HOOST, (b) the interpolation and extrapolation strategies in the HOOST.

Significantly, the HOOST specifically designs an attention block to learn sampling factors. The structure of the attention block is depicted in Fig.4. The input is passed through a squeeze operation (i.e., $\mathbf{F}_{\text{sq}}(\bullet)$) and an excitation operation (i.e., $\mathbf{F}_{\text{ex}}(\bullet)$), and is followed by a scale operation (i.e., $\mathbf{F}_{\text{scale}}(\bullet)$), which produces a sampling factor. In this paper, the squeeze operation, excitation operation, and scale operation are achieved by using a squeeze layer with the ReLU, an excitation layer with the ReLU, and an output layer with the Tanh [26], respectively. Moreover, the synthetic process of minority class samples is summarized below.

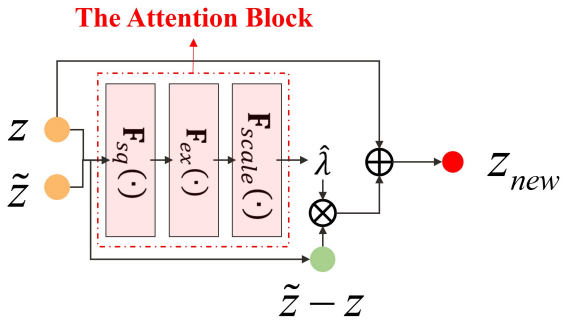


Fig.4 The building module entitled HOOST

To begin with, every minority class sample z is stochastically concatenated with one of its nearest neighbors (i.e., $\tilde{z}$) to constitute a sample pair $[z,\tilde{z}]$.

Subsequently, the attention block takes the sample pair $[z,\tilde{z}]$ as an input and outputs the sampling factor $\hat{\lambda}$. The sample pair $[z,\tilde{z}]$ is passed through a squeeze operation, and the output z_{sq} is defined as

$$z_{sq}=\mathbf{F}_{sq}\left([z,\tilde{z}]\right)=\varphi\left(W_1[z,\tilde{z}]\right)$$

where φ is the ReLU activation function, and W_1 represents parameter of the squeeze layer. Next, z_{sq} is passed through an excitation operation, and the output z_{ex} is defined as

$$z_{ex}=\mathbf{F}_{ex}\left(z_{sq}\right)=\varphi\left(W_2 z_{sq}\right)$$

where W_2 represents parameter of the excitation layer. Then, z_{ex} is followed by a scale operation, and the output $\hat{\lambda}$ is defined as

$$\hat{\lambda}=\mathbf{F}_{\text{scale}}\left(z_{ex}\right)=\sigma\times\eta\left(W_3 z_{ex}\right)+\mu$$

where η is the Tanh activation function, and W_3 represents matrix of the output layer. μ and σ are the location parameter and scale parameter, respectively, which are used to adjust the value range of the sampling factor. Obviously, the value range of $\hat{\lambda}$ is $(-1,2)$ when $\sigma=1.5,\mu=0.5$.

Third, the formula of the synthetic sample is defined as

$$z_{\text{new}}=z+\hat{\lambda}\left(\tilde{z}-z\right)$$

where z_{new} is the synthetic sample. Significantly, when $\hat{\lambda}\in(-1,0)\bigcup(1,2)$, the HOOST takes the extrapolation strategy to synthesize out-of-distribution samples; when $\hat{\lambda}\in[0,1]$, the HOOST takes the interpolation strategy to synthesize in-distribution samples, as depicted in Fig.3(b). For example, when $\hat{\lambda}=2$, $z_{\text{new}}=-z+2\times\tilde{z}=\tilde{z}+(\tilde{z}-z)$ (i.e., the synthetic sample is located at the extension line of the line segment).

Finally, the above steps are repeated until a balanced training set is obtained.

It is noteworthy that the optimization goal of the HOOST is to make synthetic samples as close as possible to the initial decision hyperplane. Based on this, this paper designs a new loss function L_{HP}, which is calculated by

$$L_{\mathrm{HP}}=\frac{1}{K}\sum_{z_i\in Z_{\mathrm{new}}}DIST\left(z_i,HP_{\mathrm{Initial}}\right)=\frac{1}{K}\sum_{z_i\in Z_{\mathrm{new}}}\frac{\left|w\bullet z_i+b\right|}{\left\|w\right\|_2}$$

where z_i is the i-th sample in the set of all synthetic samples Z_{new}, HP_{Initial} represents the initial decision hyperplane, and $DIST\left(z_i,HP_{\mathrm{Initial}}\right)$ is the distance of the synthetic sample to the initial decision hyperplane. w and b are the weight vector and bias of the initial decision hyperplane, which are obtained by the standard linear-SVM on the imbalanced training set. Moreover, K is the number of synthetic samples, $\left|\bullet\right|$ is the absolute value, and $\left\|\bullet\right\|_2$ is the 2-norm of the vector. Obviously, when minimizing the loss function L_{HP}, $DIST\left(z_i,HP_{\mathrm{Initial}}\right)$ will also decrease. That is, the synthetic sample will move towards the initial decision hyperplane along the connection line during the training process (i.e., Epoch $1\rightarrow$ Epoch $2\rightarrow\cdots\rightarrow$ Epoch E), as shown in Fig.3(a).

4.3. Pseudocode of the HOOST

The procedure of the HOOST is explained briefly, which is given as follows. The input of the algorithm is the training set $\{Z,Y\}$ (Z is the set of samples in the training set, and Y is the set of labels in the training set; the label of the real or synthetic minority class sample is 1, and the label of majority class sample is 0) and the number of synthetic minority class samples K, and the output is the balanced training set $\left\{\left[Z,Z_{\mathrm{new}}\right],\left[Y,Y_{\mathrm{new}}\right]\right\}$.

Algorithm 1: HOOST Algorithm

Input: Training set $\{Z,Y\}$, the number of synthetic minority class samples K

Output: Balanced training set $\left\{\left[Z,Z_{\mathrm{new}}\right],\left[Y,Y_{\mathrm{new}}\right]\right\}$

1 Let $Z_{\mathrm{H}}\leftarrow$ the healthy set; $Z_{\mathrm{F}}\leftarrow$ the fault set; $Z_{\mathrm{new}}\leftarrow$ an empty set; $Y_{\mathrm{new}}\leftarrow$ an empty set; $N_{\mathrm{new}}\leftarrow$ the number of samples in the set Z_{new}; $N_{\mathrm{F}}\leftarrow$ the number of samples in the set Z_{F}; $k\leftarrow$ the number of nearest neighbors (5 by default)

2 **while** $N_{\mathrm{new}}\le K$ **do**

3 **for** $i\leftarrow 1$ **to** N_{F} **do**

$z \leftarrow Z_{\mathrm{F}}[i]$

$\tilde{z} \leftarrow$ random one of k nearest neighbors of z

$\hat{\lambda} \leftarrow Attention_Block([z, \tilde{z}])$

$z_{\mathrm{new}} \leftarrow z + \hat{\lambda}(\tilde{z} - z)$

$Z_{\mathrm{new}} \leftarrow Z_{\mathrm{new}} \bigcup \{z_{\mathrm{new}}\}$

$Y_{\mathrm{new}} \leftarrow Y_{\mathrm{new}} \bigcup \{1\}$

if $N_{\mathrm{new}} > K$ **do**

break

end if

end for

end while

return $\{[Z, Z_{\mathrm{new}}], [Y, Y_{\mathrm{new}}]\}$

5. Proposed imbalanced fault detection framework

In this section, we illustrate the classical multi-head self-attention network (MHSANet) and summarize the general procedure of the proposed fault detection framework.

5.1. The classical MHSANet

Recently, the MHSANet and its variants have shown a strong ability to process time-series data, especially for the extraction of complex temporal patterns in monitoring data [27], [28]. The classical MHSANet consists of multiple Transformer Encoder layers (i.e., a multi-head self-attention mechanism, two short-cut & layer normalization, and a fully connected feed-forward layer) and a linear output layer with the Softmax. The core component is the multi-head self-attention mechanism, as illustrated in Fig.5.

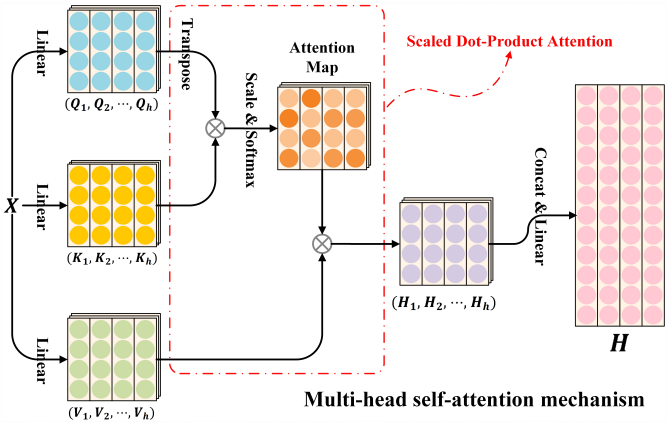


Fig.5 The architecture of multi-head self-attention mechanism.

The multi-head self-attention mechanism is formed by merging multiple parallel self-attention submodules [29]. First, the input X is passed through the three linear projections in the $i-\text{th}$ $i\text{th}$ self-attention submodule, which is calculated by

$$Q_i = XW_i^Q, K_i = XW_i^K, V_i = XW_i^V$$

where Q_i, K_i, V_i are three self-attention matrices, and W_i^Q, W_i^K, W_i^V are parameters of three linear layers. Then, the output of the $i-\text{th}$ self-attention submodule H_i is calculated by

$$H_i = \text{Attention_Map}(Q_i, K_i)V_i = \text{softmax}\left(\frac{Q_i K_i^T}{\sqrt{d_i}}\right)V_i$$

where d_i is the dimension of Q_i. Finally, the information obtained by the multi-head self-attention mechanism H is as follow:

$$H = \text{concat}\left(H_1, H_2, \cdots, H_h\right)W$$

where h is the number of self-attention sub-modules, and W is the parameter of the linear layer.

5.2. Space mapping for better inter-class separability

In practice, minority class samples highly overlap with majority class samples. Conventional over-sampling techniques face the challenge of synthesizing noisy samples, which reduce the diagnostic accuracy of the classifiers. Aiming at this issue, a self-attention Encoder-Decoder network is used to learn a feature projection, which maps the original data into a well-separable feature space, in which samples of different categories are far away from each other. In this situation, synthetic samples have a high probability of falling into the minority class region and reduce the possibility of synthesizing noisy samples. Moreover, the Encoder network consists of multiple Transformer Encoder layers, while the Decoder network consists of multiple Transformer Encoder layers and a linear output layer.

For the Encoder-Decoder network, there are two optimization goals: (1) the feature representation of input data can retain important information; (2) the feature representation of different categories of data is more linearly separable. In view of this, this paper combines the reconstruction loss and triplet-center loss to simultaneously train Encoder-Decoder network.

To meet the first goal, this paper uses class-balanced $l_2-\text{norm}$ as the reconstruction loss L_{CR} [30], which is as follow:

$$L_{\text{CR}} = \frac{1}{|X|}\sum_{x_i \in X}\frac{1-\beta}{1-\beta^{n_i}}\left\|x_i - \hat{x}_i\right\|_2$$

where x_i is the $i-\text{th}$ sample in the set X, and $\hat{x}_i$ is the reconstructed representation of x_i.

$|X|$ is the number of samples in the set X, and $\|\bullet\|_2$ is the l_2-norm. β is a hyper-parameter and is set to 0.999 [30]. Moreover, if x_i is a majority class sample, n_i is the number of minority class samples; otherwise, n_i is the number of majority class samples.

Meanwhile, to meet the second goal, triplet-center loss is selected [31], which is calculated by

$$L_{\mathrm{TC}}=\frac{1}{|Z|}\sum_{z_i\in Z}\max\left(\|z_i-c_i\|_2+d-\|z_i-\tilde{c}_i\|_2,0\right)$$

where z_i is the $i-$th sample in the set Z, $|Z|$ is the number of samples in the set Z. d is a hyperparameter and is set to 5 [31]. In addition, if z_i is a majority class sample, c_i is the center of majority class samples, and $\tilde{c}_i$ is the center of minority class samples; otherwise, c_i is the center of minority class samples, and $\tilde{c}_i$ is the center of majority class samples.

Therefore, the final loss function is calculated by

$$L_{\mathrm{ED}}=L_{\mathrm{CR}}+L_{\mathrm{TC}}$$

5.3. General procedure of the fault detection framework

To tackle the issues caused by class imbalance, the developed HOOST is put together with a clustering-based under-sampling technique to serve as a hybrid re-sampling scheme. Furthermore, as depicted in Fig.6, the proposed fault detection framework consists of three relatively independent modules, namely (I) under-sampling module; (II) hyperplane-oriented over-sampling module; (III) fault classification module.

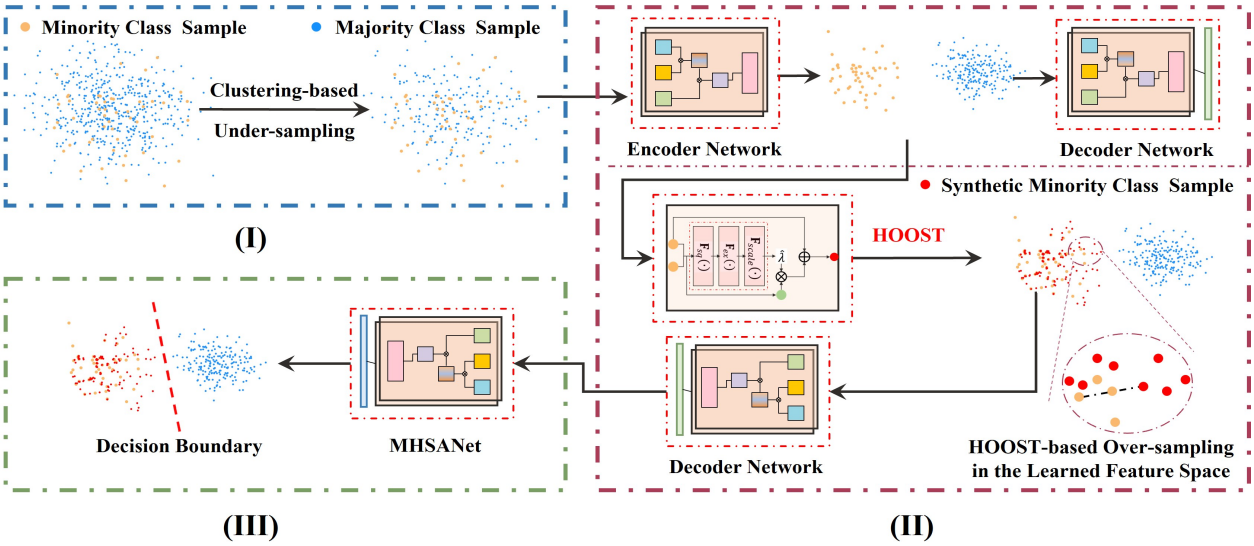


Fig.6 The procedures of the developed fault detection framework.

The procedures of the proposed framework are summarized as follows. Without loss of generality, let X_{MA} be a set of N_{MA} majority class samples and X_{MI} a set of N_{MI} minority class samples in the training dataset.

(I) Under-sampling module: The under-sampling module adopts the unsupervised clustering method to divide majority class samples X_{MA} into different clusters $C^{*}=\left\{c_{1}^{*},c_{2}^{*},\cdots,c_{K}^{*}\right\}$, where c_{j}^{*} is the center of the $j-\mathrm{th}$ cluster. Then, C^{*} is used to replace X_{MA} and is merged with X_{MI} to form the under-sampled training set X_{U} (i.e., $X_{\mathrm{U}}=C^{*}\bigcup X_{\mathrm{MI}}$). It aims to reduce the quantity difference between different categories. This paper adopts the simple and effective K-Means algorithm as the aforementioned unsupervised clustering method, which can efficiently find the relatively optimal cluster division, and the cluster center can also better represent the data in the whole cluster.

(II) Hyperplane-oriented over-sampling module: The hyperplane-oriented over-sampling module maps X_{U} into a linearly separable feature space using the Encoder network (i.e., $Z=\mathrm{Encoder}\left(X_{\mathrm{U}}\right)$), synthesizes minority class samples using the developed HOOST (i.e., $Z_{\mathrm{new}}=\mathrm{HOOST}\left(Z\right)$), and maps synthetic samples back to the original space using the Encoder network (i.e., $\hat{X}_{\mathrm{new}}=\mathrm{Decoder}\left(Z_{\mathrm{new}}\right)$). Then, the set $\hat{X}_{\mathrm{new}}$ and X_{U} together form the balanced training set X_{B} (i.e., $X_{\mathrm{B}}=X_{\mathrm{U}}\bigcup\hat{X}_{\mathrm{new}}$). This module can not only reduce the possibility of synthesizing noisy samples, but also extend the distribution region occupied by minority class samples.

(III) Fault classification module: The fault classification module is used to learn a decision boundary on the X_{B} under the guidance of cross-entropy loss, and apply the decision boundary to detect the health condition of aero-engines (i.e., healthy or fault). As described in Section 5.1, the MHSANet is especially suitable for temporal monitoring data of aero-engines, which is selected to learn the final decision boundary.

6. Experimental Results

6.1. Experimental setup

6.1.1. Dataset descriptions

The real-world monitoring data comes from the CFM56-7B turbofan engines of an Asian airline. Fig.7(a) shows the structure and sensor arrangement of the target aero-engine, which is primarily made up of the fan, low-pressure compressor (LPC), high-pressure compressor (HPC), combustion chamber (CC), high-pressure turbine (HPT), low-pressure turbine (LPT), etc. To monitor the health status of aero-engines, many sensors are installed on these modules to measure the aero-engine performance parameters, including exhaust gas temperature (EGT), low-pressure rotor speed (N1), high-pressure rotor speed (N2), fuel flow (FF), etc.

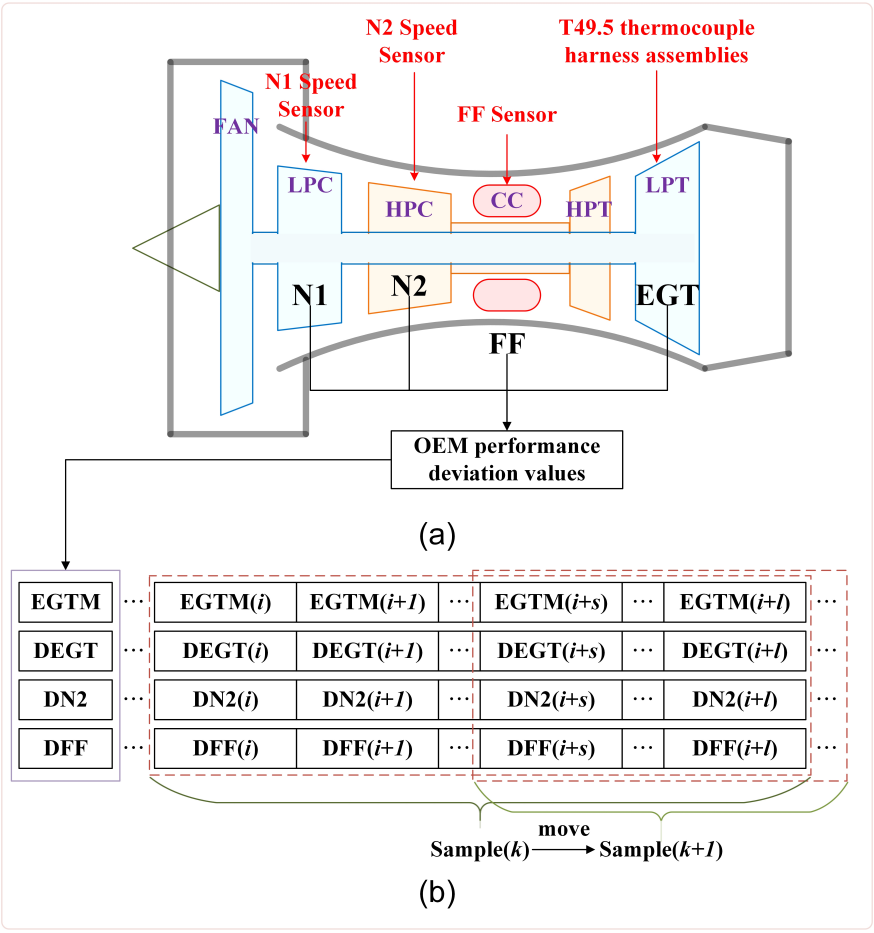


Fig.7 (a) Acquisition and preprocessing flow of aero-engine performance data, (b) sample construction using a sliding window method, where $\mathrm{Sample}(k)$ denotes the $k-$th sample.

However, measurements of these parameters are affected by operating conditions, external environment, and other factors. To monitor the health status of aero-engines more accurately, the original equipment manufacturer (OEM) converts them into exhaust gas temperature margin (EGTM), deviation of exhaust gas temperature (DEGT), deviation of high-pressure rotor speed (DN2), and deviation of fuel flow (DFF) [32], which are also selected as key performance parameters of aero-engines in this paper. According to the maintenance report of the airline, the monitoring data of 63 aero-engines were collected before disassembly.

In addition, as illustrated in Fig.7(b), the sliding window method is adopted to construct samples (window size $l=10$, step length $s=10$). And the five-fold cross-validation is conducted to split the data set. A total of 1017 samples are obtained, including 954 healthy samples and 63 fault samples, and the specific quantity distribution is summarized in Table I.

TABLE I. Summary of the aero-engine dataset

Five-fold cross-validation	Number of training samples		Number of test samples	
	Healthy	Fault	Healthy	Fault
1st-Fold	763	50	191	13
2nd-Fold	763	50	191	13
3rd-Fold	763	50	191	13
4th-Fold	763	51	191	12
5th-Fold	764	51	190	12

6.1.2. Comparative Re-sampling techniques

This developed method is compared with not only 1 baseline method but also 8 popular re-sampling techniques, which are detailed below.

(I) Baseline method: The baseline method does not use any re-sampling technique to balance the training set; that is, the classifier is trained directly on the imbalanced training set.

(II) 8 re-sampling techniques: These techniques are used to balance the training set, and the classifier is trained on the balanced dataset. The 8 re-sampling techniques can be classified into over-sampling and hybrid re-sampling techniques. Over-sampling techniques include SMOTE [16], Borderline Over-sampling (BOS) [33], Safe-level SMOTE (SL-SMOTE) [22], and KMeans-SMOTE (KM-SMOTE) [20]. Hybrid re-sampling techniques include SMOTE + Tomek links (SMOTE-TL) [34], SMOTE + Edited Nearest Neighbor Rule (SMOTE-ENN) [34], SMOTE + Iterative-Partitioning Filter (SMOTE-IPF) [35], and Versatile Improved SMOTE + Rough Sets (VIS-RST) [36].

6.1.3. Evaluation metrics

This paper adopts the recall rate of fault samples (ROF), recall rate of healthy samples (ROH), and class average accuracy (CAA) to verify the fault detection ability of the framework comprehensively. The above metrics are defined as

$$ROF = N_{11} / \left(N_{11} + N_{12} \right)$$

$$ROH = N_{22} / \left(N_{21} + N_{22} \right)$$

$$CAA = \left(ROF + ROH \right) / 2$$

where $N_{11}, N_{12}, N_{21},$ and N_{22} represent the number of fault samples classified as fault, that of fault samples classified as healthy, that of healthy samples classified as fault, and that healthy classified as healthy, respectively, which are summarized in Table II. Obviously, the above metrics will not be biased to majority class samples, that is, they can still evaluate the proposed framework under the condition of class imbalance.

TABLE II. An introduction of the confusion matrix

Actual Class	Predicted Class	
	Fault	Healthy
Fault	N_{11}	N_{12}
Healthy	N_{21}	N_{22}

6.2. Key hyperparameters setup

Experiments are carried out using the Python 3.9 with PyTorch 1.11.0 and an Intel Core

i5-12400F with 16G DDR4 processor. The hyper-parameters are configured according to the [32]. First, the cluster number in the under-sampling module is set to 100. Second, the Encoder network consists of two Transformer Encoder layers, in which the number of self-attention sub-modules h is set to 4; the dimension of features d_i is set to 128. Third, the Decoder network is also composed of two Transformer Encoder layers and a linear output layer, whose hyper-parameters are the same as that of the Encoder network. Fourth, the HOOST-related hyper-parameters are the dimension of the squeeze linear layer and the dimension of the excitation linear layer, which are set to 160, 2560, respectively. Fifth, the classification network consists of two Transformer Encoder layers and a linear output layer, whose hyper-parameters are the same as that of the Encoder network. Moreover, this paper adopts the Adam to train the weight parameters of the neural network, where the learning rate is 0.0002, the first momentum is 0.5 and the second momentum is 0.999, and the batch size is set to 128.

6.3. Validating the effectiveness of the developed method

Figs.8-9 show the ROF and ROH of the considered fault detection methods. First, the ROF is used to compare the ability of different methods to diagnose fault samples. It is clear that, for fault samples, the diagnostic ability of the developed method is superior to that of the baseline method and 8 comparison re-sampling techniques in all five-fold experiments. Second, the ROH is used to compare the ability of different methods to diagnose healthy samples. The diagnostic ability of the developed method is not as good as that of the Baseline method (the first place in terms of ROH). This is due to the fact that the Baseline method is more biased towards healthy samples and ignores fault samples. On the whole, the developed method is the most promising candidate among the considered methods for aero-engines with highly severe safety criteria.

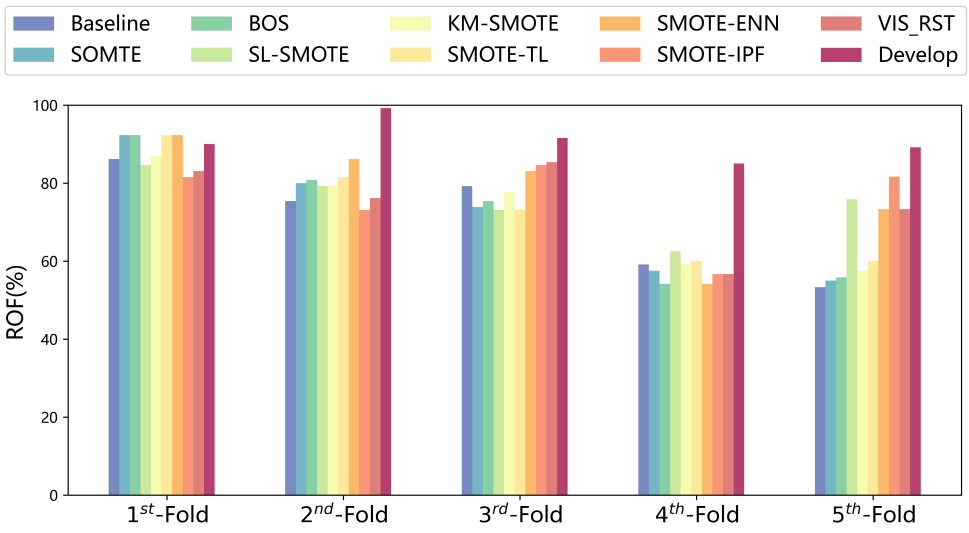


Fig.8 ROF of different fault detection models.

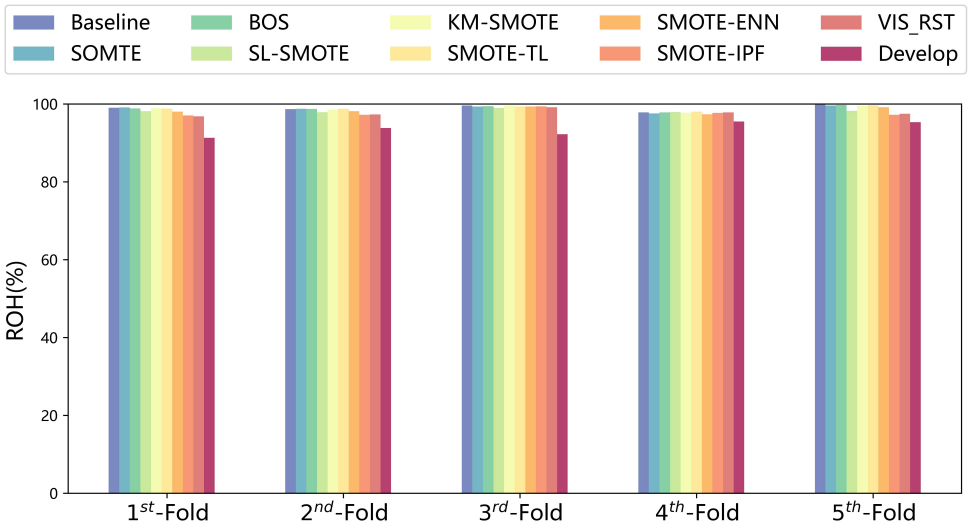


Fig.9 ROH of different fault detection models.

Furthermore, the average values of ROF, ROH and CAA are summarized in Table III, and the main conclusions are drawn below.

TABLE III. Average ROF, ROH, CAA of different fault detection methods

Category	Method	ROF (%)	ROH (%)	CAA (%)
None	Baseline	70.65	**98.96**	84.80
Over-sampling	SOMTE	71.73	98.85	85.29
	BOS	71.69	98.90	85.29
	SL-SMOTE	75.05	98.20	86.62
	KM-SMOTE	72.10	98.83	85.46
Hybrid re-sampling	SMOTE-TL	73.38	98.90	86.14
	SMOTE-ENN	77.80	98.38	88.09
	SMOTE-IPF	75.51	97.70	86.60
	VIS-RST	74.92	97.70	86.31
our	**Develop**	**90.98**	93.62	**92.30**

(1) In terms of ROF and CAA, all re-sampling techniques are superior to the Baseline method. And 8 re-sampling techniques are similar to the Baseline method for ROH. It is clear that the suitable re-sampling technique can enhance the classifier's diagnostic ability on the imbalanced dataset.

(2) In terms of ROF and CAA, both SL-SMOTE and KM-SMOTE outperform the SMOTE. Moreover, for ROH, they are also very close to the SMOTE, with their differences smaller than 0.76%. As described in Section 2, SL-SMOTE and KM-SMOTE are two variants of SMOTE that can reduce the possibility of synthesizing noisy samples, so that their performance is improved.

(3) In terms of ROF and CAA, the SMOTE-TL, SMOTE-ENN, and SMOTE-IPF are superior to the SMOTE. They belong to hybrid re-sampling techniques (SMOTE+Under-sampling), which show the advantages of hybrid re-sampling techniques.

(4) The developed method yields the highest diagnostic ability compared to the considered comparison methods in terms of ROF and CAA. To be specific, compared with the hybrid

re-sampling techniques (i.e., SMOTE-TL, SMOTE-ENN, SMOTE-IPF, and VIS-RST), its CAA is increased by 6.16%, 4.21%, 5.70%, and 5.99%, respectively. In terms of ROF, it is increased by 17.60%, 13.18%, 15.47%, and 16.06%, respectively. It shows that our hybrid re-sampling scheme is very effective.

6.4. Investigation on performance with different numbers of clusters

Tables IV-VI show the ROF, ROH, and CAA of the developed method under the condition of different numbers of cluster centers, respectively. In terms of average ROF, ROH, and CAA, the developed method yields the highest diagnostic ability when the number of cluster centers is 70, 150, and 100, respectively. On the whole, with the increase of the number of cluster centers, the diagnostic ability of the developed method gradually became stronger for healthy samples while weaker for fault samples. When the number of cluster centers is 100, the developed method yields the highest comprehensive diagnostic ability. Experimental results indicate that the proper number of cluster centers is significant for the performance of the developed method. Moreover, when the number of cluster centers is 50, the developed method becomes a cluster-based under-sampling technique, and the performance is the worst in terms of CAA, further demonstrating the effectiveness of the developed method.

TABLE IV. ROF (%) when using different value for selecting clusters

K	1st-Fold	2nd-Fold	3rd-Fold	4th-Fold	5th-Fold	Average
50	92.30	100	89.23	76.66	95.00	90.64
60	93.07	99.23	91.53	76.66	94.16	90.93
70	95.38	99.23	92.30	77.50	95.00	**91.88**
80	88.46	100	83.84	83.33	88.33	88.79
90	92.30	99.23	92.30	80.83	87.50	90.43
100	90.00	99.23	91.53	85.00	89.16	90.98
110	92.30	97.69	95.38	79.16	86.66	90.24
120	92.30	96.15	87.69	76.66	88.33	88.23
130	90.76	100	88.46	81.66	78.33	87.84
140	90.76	99.23	93.07	78.33	75.00	87.28
150	90.76	98.46	86.15	77.50	69.16	84.41

TABLE V. ROH (%) when using different value for selecting clusters

K	1st-Fold	2nd-Fold	3rd-Fold	4th-Fold	5th-Fold	Average
50	90.41	90.68	87.74	92.51	84.52	89.17
60	91.93	91.72	91.20	94.86	92.31	92.41
70	92.04	90.00	92.09	94.60	89.36	91.62
80	93.92	93.87	91.93	95.96	93.36	93.81
90	90.47	94.18	94.50	93.97	95.00	93.62
100	91.30	93.82	92.19	95.49	95.31	93.62
110	92.51	92.30	94.71	94.45	95.57	93.91
120	92.82	94.92	94.13	95.60	94.94	94.48
130	93.29	96.64	94.29	95.91	96.26	95.28
140	94.65	95.39	93.97	95.91	95.78	95.14
150	94.97	95.28	95.07	95.54	98.00	**95.77**

TABLE VI. CCA (%) when using different value for selecting clusters

K	1st-Fold	2nd-Fold	3rd-Fold	4th-Fold	5th-Fold	Average
50	91.36	95.34	88.48	84.58	89.76	89.90
60	92.50	95.47	91.37	85.76	93.24	91.67
70	93.71	94.61	92.20	86.05	92.18	91.75
80	91.19	96.93	87.89	89.65	90.85	91.30
90	91.38	96.70	93.40	87.40	91.25	92.03
100	90.65	96.52	91.86	90.24	92.24	**92.30**
110	92.41	94.99	95.04	86.80	91.12	92.07
120	92.56	95.53	90.91	86.13	91.64	91.35
130	92.03	98.32	91.37	88.79	87.29	91.56
140	92.71	97.31	93.52	87.12	85.39	91.21
150	92.87	96.87	90.61	86.52	83.58	90.09

6.5. Investigation on the simulation data

In this subsection, to further verify the HOOST, this paper adopts the simulation data that is generated using the two-dimensional Gaussian distribution. Assuming that majority class samples obey the Gaussian distribution $f_1(X;\mu_1,\Sigma_1)$ and minority class samples obey the Gaussian distribution $f_2(X;\mu_2,\Sigma_2)$. More specifically, the probability density function (PDF) of majority class samples is defined below.

$$f_1(X;\mu_1,\Sigma_1)=\frac{1}{2\pi\sqrt{|\Sigma_1|}}e^{-\frac{(X-\mu_1)^T\Sigma_1^{-1}(X-\mu_1)}{2}}$$

where $\mu_1=[3,0]$ and $\Sigma_1=\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$ represent the mean and covariance of the Gaussian distribution f_1. And the PDF of minority class samples is defined as

$$f_2(X;\mu_2,\Sigma_2)=\frac{1}{2\pi\sqrt{|\Sigma_2|}}e^{-\frac{(X-\mu_2)^T\Sigma_2^{-1}(X-\mu_2)}{2}}$$

where $\mu_2=[-3,0]$ and $\Sigma_2=\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$ represent the mean and covariance of the Gaussian distribution f_2. In this paper, we obtain an imbalanced dataset, including 200 majority class samples and 50 minority class samples, as shown in Fig.10(a).

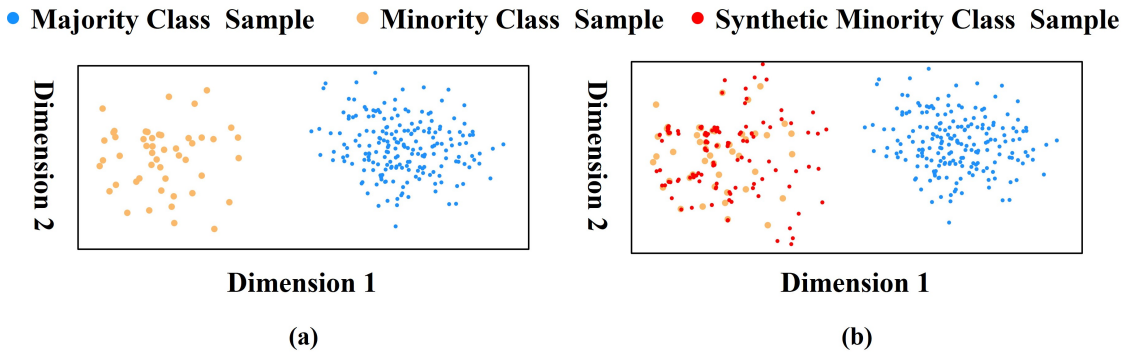


Fig.10 2D sketch maps of (a) simulation data; (b) simulation data after HOOST-based over-sampling in the original data space.

Furthermore, Fig.10(b) shows that the distribution of simulation samples after HOOST. It is clear that the HOOST can generate not only in-distribution but also out-of-distribution synthetic data close to the safety distribution region around the decision hyperplane. The synthetic data effectively expands the distribution region occupied by minority class samples, which provides favorable circumstances for the estimation of the decision boundary.

7. Conclusion

To tackle the class imbalance and overlap, a hyperplane-oriented over-sampling technique, namely HOOST, is developed. To be specific, the HOOST not only adopts the interpolation strategy to synthesize in-distribution samples, but also adopts the extrapolation strategy to synthesize out-of-distribution samples. Furthermore, to decrease the possibility of synthesizing noisy samples, a self-attention Encoder-Decoder is applied to learn a well-separable feature space, in which the overlap of samples from different categories is weakened. Finally, the developed HOOST is integrated with a clustering-based under-sampling technique, in order to better re-balance the data distribution.

The developed method is demonstrated through a series of experiments on an actual aero-engine dataset. More specifically, the developed method has increased by 6.16%, 4.21%, 5.70%, and 5.99% in terms of average CAA compared with SMOTE-TL, SMOTE-ENN, SMOTE-IPF, and VIS-RST. The reasons are that, on one hand, the developed method can retain more valuable information; on the other hand, the HOOST reduces the possibility of synthesizing noisy samples and extends the region occupied by minority class samples.

Finally, fault detection of aero-engines is an evolving research topic. In the future, we will focus on developing novel re-sampling techniques to further improve fault detection accuracies of aero-engines under various operating conditions.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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